Vlookup and Hlookup Functions

The Vlookup (Vertical lookup) function looks for a value in the leftmost column of a table, and then returns a value in the same row from another column you specify.

Steps:

1. Insert the Vlookup function in another table. A function argument dialog box appears.
2. Enter a value for the Lookup_value field. This field contains the value that is to be found in the first column of the main table, and can be a value, a reference, or a text string.
3. Next, select a set of values for table_array field that contains data to be retrieved. It can be a reference or a range.
4. Then, provide a Col_index_num value. It is the column number in table_array from which the matching value should be returned.
5. Lastly, in the range_lookup field, provide a Boolean value. The boolean False tells the Vlookup function to return an exact match. If the VLOOKUP function cannot find the value in the first column, it will return a #N/A error. The boolean TRUE tells the VLOOKUP function to return an approximate match. If the VLOOKUP function cannot find the value in the first column, it will return the largest value smaller than that number.

If the leftmost column of the table contains duplicates, the Vlookup function matches the first instance only. Also, it performs case-insensitive lookups.

In a similar way, you can use the Hlookup (Horizontal lookup) function.

VLOOKUP and HLOOKUP are functions in Excel that allow users to search a table of data based on what the user has supplied and give appropriate information from that table. *VLOOKUP* allows searching in a table that is set up vertically. All of the data should be set up in columns and each column is responsible for one kind of data. *HLOOKUP* is the exact same function but looks up data that has been formatted by rows instead of columns.
VLOOKUP/HLOOKUP have three arguments in order.

lookup_value	User input. This is the value that the function uses to search on.
table_array	The area of cells in which the table is located. This includes the entire range/table from which the data will be extracted. All columns of the data must be included.
col_index_num	The column of data that contains the answer. For *HLOOKUP*, this will be *row_index_num*.
range_lookup	A TRUE or FALSE value. When set to TRUE, the lookup function gives the closest match to the *lookup_value* without going over the *lookup_value*. When set to FALSE, an exact match must be found to the *lookup_value* or the function will return *#N/A*. Note, this requires that the column containing the *lookup_value* be formatted in ascending order.

Formula Errors

Some of the common Excel formula errors are as follows:

Error	Description	Solution
##### Error	When your cell contains this error code, the column isn't wide enough to display the value.	Click on the right border of the column header and increase the column width or Double click the right border of the column header to automatically fit the widest entry in that column.
#NAME? error	It occurs when Excel does not recognize text in a formula.	Correct the spelling mistake.
#VALUE! Error	Excel displays this error when a formula has the wrong type of argument.	Either change the value of cell to a number or Use a function to ignore cells that contain text.
#DIV/0! Error	Excel displays this error when a formula tries to divide a number by 0 or an empty cell.	Either change the value of the cell (in denominator) to a value that is not equal to 0 or Prevent the error from being displayed by using the logical function IF.
#REF! error	Excel displays this error when a formula refers to a cell that is not valid (or deleted afterwards).	To fix this error, you can either delete +#REF! in the formula of the cell or you can undo your deletion.

Lab Session 06

OBJECT

Working with Formulas conditions, logical &Trigonometric functions in Microsoft Excel

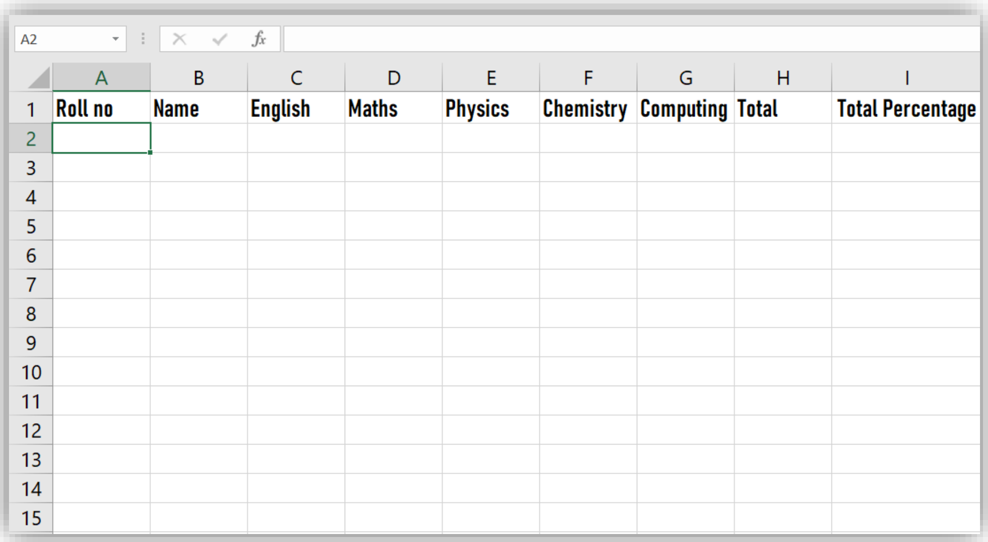
Entering Formulas

EXERCISES

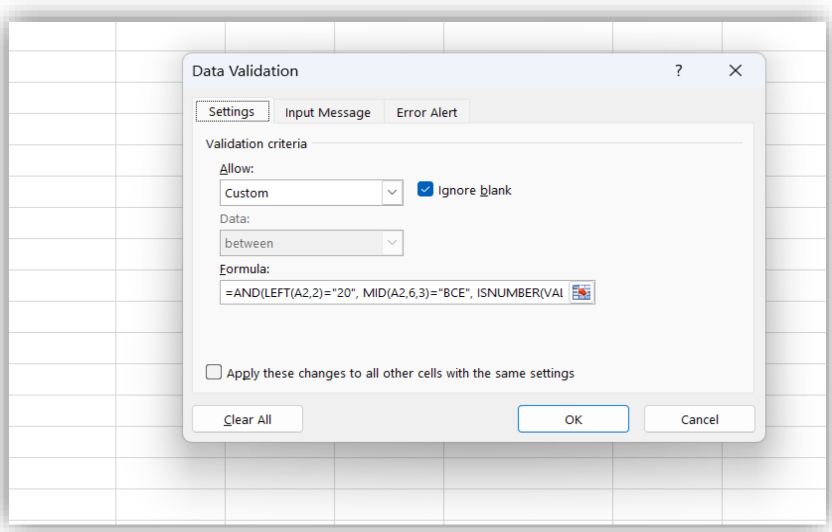
Task 01

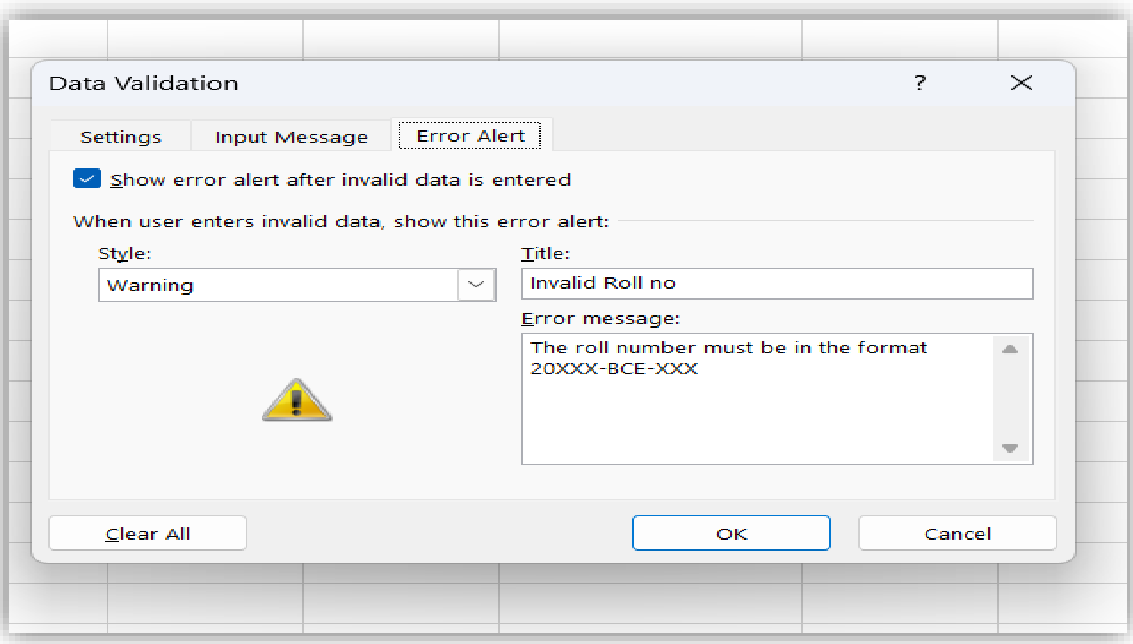
Perform the following operations then print and attach your result.

1. Make the following columns in your worksheet: Roll No, Name, English, Math, Physics, Chemistry, Computing, Total, Percentage. Resize the columns if required.



2. Apply data validation to ensure that the roll number is of the format 20XXX-BCE-XXX.

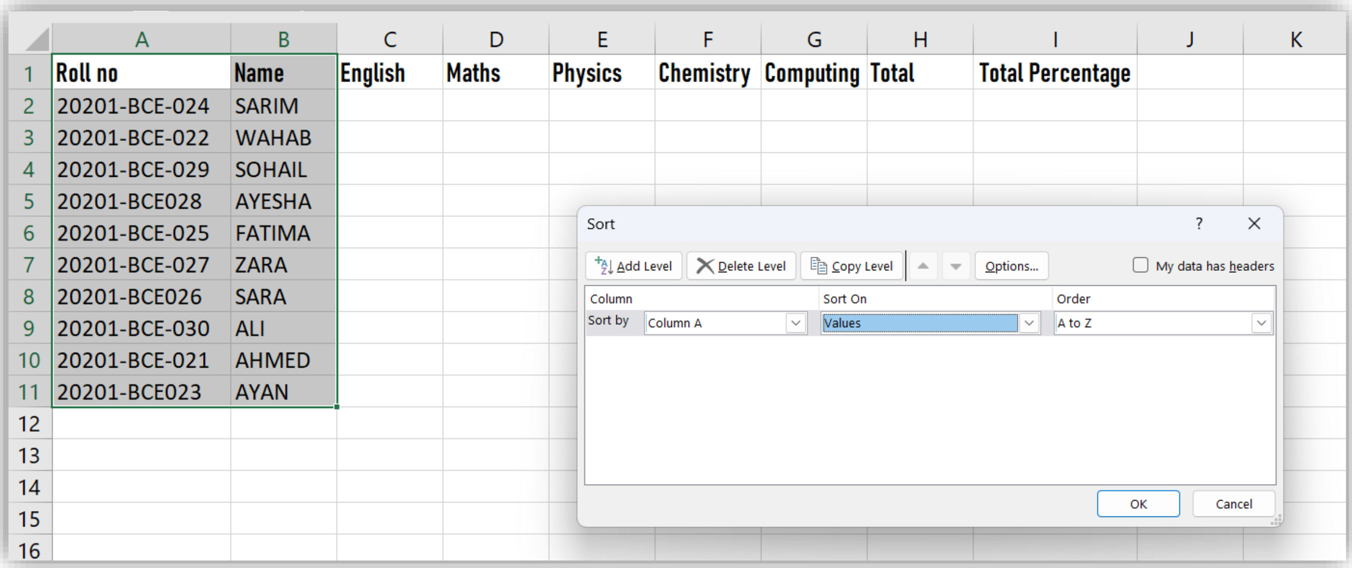




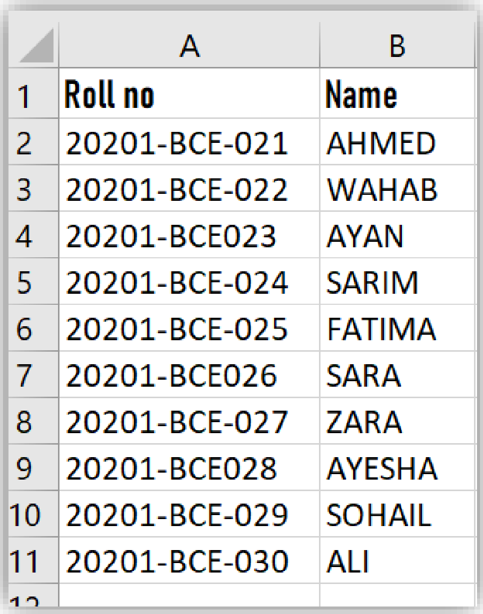
3. Enter random roll numbers and student names (at least 10).

	A	B
1	**Roll no**	**Name**
2	20201-BCE-024	SARIM
3	20201-BCE-022	WAHAB
4	20201-BCE-029	SOHAIL
5	20201-BCE028	AYESHA
6	20201-BCE-025	FATIMA
7	20201-BCE-027	ZARA
8	20201-BCE026	SARA
9	20201-BCE-030	ALI
10	20201-BCE-021	AHMED
11	20201-BCE023	AYAN

4. Sort the students based on their roll numbers.



	A	B	C	D	E	F	G	H	I	J	K
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage		
2	20201-BCE-024	SARIM									
3	20201-BCE-022	WAHAB									
4	20201-BCE-029	SOHAIL									
5	20201-BCE028	AYESHA									
6	20201-BCE-025	FATIMA									
7	20201-BCE-027	ZARA									
8	20201-BCE026	SARA									
9	20201-BCE-030	ALI									
10	20201-BCE-021	AHMED									
11	20201-BCE023	AYAN									
12											
13											
14											
15											
16											



	A	B
1	Roll no	Name
2	20201-BCE-021	AHMED
3	20201-BCE-022	WAHAB
4	20201-BCE023	AYAN
5	20201-BCE-024	SARIM
6	20201-BCE-025	FATIMA
7	20201-BCE026	SARA
8	20201-BCE-027	ZARA
9	20201-BCE028	AYESHA
10	20201-BCE-029	SOHAIL
11	20201-BCE-030	ALI

5. Use the *RANDBETWEEN* function to enter the marks of each subject (0-100).

	A	B	C	D	E	F	G
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing
2	20201-BCE-021	AHMED	=RANDBETWEEN(0, 100)				
3	20201-BCE-022	WAHAB					
4	20201-BCE023	AYAN					
5	20201-BCE-024	SARIM					
6	20201-BCE-025	FATIMA					
7	20201-BCE026	SARA					
8	20201-BCE-027	ZARA					
9	20201-BCE028	AYESHA					
10	20201-BCE-029	SOHAIL					
11	20201-BCE-030	ALI					

	A	B	C	D	E	F	G	H	I
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage
2	20201-BCE-021	AHMED	44	44	75	71	66		
3	20201-BCE-022	WAHAB	94	6	48	41	4		
4	20201-BCE023	AYAN	35	64	51	94	18		
5	20201-BCE-024	SARIM	70	49	100	53	34		
6	20201-BCE-025	FATIMA	3	21	23	79	31		
7	20201-BCE026	SARA	51	92	71	74	12		
8	20201-BCE-027	ZARA	14	45	61	8	32		
9	20201-BCE028	AYESHA	52	97	12	16	22		
10	20201-BCE-029	SOHAIL	90	39	79	46	1		
11	20201-BCE-030	ALI	82	89	58	69	31		

6. Calculate the total marks of a student using the *SUM* function.

H2 =SUM(C2:G2)

	A	B	C	D	E	F	G	H	I
1	Roll no	Name	English	Maths	Physics	Chemistry	Computing	Total	Total Percentage
2	20201-BCE-021	AHMED	86	33	36	17	57	229	
3	20201-BCE-022	WAHAB	37	65	60	89	46	297	
4	20201-BCE023	AYAN	31	25	66	32	86	240	
5	20201-BCE-024	SARIM	61	12	100	37	84	294	
6	20201-BCE-025	FATIMA	9	73	27	76	56	241	
7	20201-BCE026	SARA	69	3	78	64	35	249	
8	20201-BCE-027	ZARA	61	37	69	82	38	287	
9	20201-BCE028	AYESHA	40	20	53	27	76	216	
10	20201-BCE-029	SOHAIL	39	1	52	77	96	265	
11	20201-BCE-030	ALI	3	37	58	27	34	159	

7. Calculate the percentage of each student by entering the appropriate formula. Round it to 1 decimal place.

	A	B	C	D	E	F	G	H	I
1	**Roll no**	**Name**	**English**	**Maths**	**Physics**	**Chemistry**	**Computing**	**Total**	**Total Percentage**
2	20201-BCE-021	AHMED	65	70	50	15	59	259	51.8
3	20201-BCE-022	WAHAB	97	100	92	37	1	327	
4	20201-BCE023	AYAN	10	89	96	9	29	233	
5	20201-BCE-024	SARIM	27	75	88	75	95	360	
6	20201-BCE-025	FATIMA	29	53	84	93	53	312	
7	20201-BCE026	SARA	1	14	32	13	81	141	
8	20201-BCE-027	ZARA	74	74	91	71	28	338	
9	20201-BCE028	AYESHA	29	95	60	98	79	361	
10	20201-BCE-029	SOHAIL	34	100	94	65	53	346	
11	20201-BCE-030	ALI	73	51	44	83	49	300	

	A	B	C	D	E	F	G	H	I
1	**Roll no**	**Name**	**English**	**Maths**	**Physics**	**Chemistry**	**Computing**	**Total**	**Total Percentage**
2	20201-BCE-021	AHMED	54	18	84	33	85	274	54.8
3	20201-BCE-022	WAHAB	65	45	84	82	77	353	70.6
4	20201-BCE023	AYAN	59	32	3	64	31	189	37.8
5	20201-BCE-024	SARIM	96	43	41	25	70	275	55
6	20201-BCE-025	FATIMA	23	18	1	71	74	187	37.4
7	20201-BCE026	SARA	73	82	80	3	16	254	50.8
8	20201-BCE-027	ZARA	59	21	53	76	58	267	53.4
9	20201-BCE028	AYESHA	85	84	3	72	56	300	60
10	20201-BCE-029	SOHAIL	63	7	60	5	24	159	31.8
11	20201-BCE-030	ALI	29	51	74	98	48	300	60

8. Find the maximum and minimum marks of English. Then use the fill handle to find the maximum and minimum marks of all the subjects.

	A	B	C	D	E	F	G	H	I
1	**Roll no**	**Name**	**English**	**Maths**	**Physics**	**Chemistry**	**Computing**	**Total**	**Total Percentage**
2	20201-BCE-021	AHMED	15	35	89	35	33	207	41.4
3	20201-BCE-022	WAHAB	15	44	44	56	87	246	49.2
4	20201-BCE023	AYAN	9	92	13	31	28	173	34.6
5	20201-BCE-024	SARIM	39	60	45	78	50	272	54.4
6	20201-BCE-025	FATIMA	20	64	93	0	98	275	55
7	20201-BCE026	SARA	16	18	81	66	22	203	40.6
8	20201-BCE-027	ZARA	84	40	58	36	58	276	55.2
9	20201-BCE028	AYESHA	78	61	32	87	99	357	71.4
10	20201-BCE-029	SOHAIL	54	20	35	8	29	146	29.2
11	20201-BCE-030	ALI	53	96	86	58	2	295	59
12	**Max. marks of subject**		84	96	93	87	99		
13	**Min. marks of subjects**		9	18	13	0	2		

9. Use a formula to automatically assign grades based on the following criteria: A: 90-100%, B: 80-90%, C: 70-80%, F: < 70%. (Use logical IF).

10. Apply the following conditional formatting to the data: A: Green, B: Blue, C: Yellow, D: Red.

11. Perform *VLOOKUP* on your data. Set the roll number as the lookup field. The data returned should be the percentage

12. Design a TEMPERATURE CONVERTER from Celsius to Fahrenheit and Fahrenheit to Celsius. Present the Converter with appropriate design. Formulas for conversion are as follows:

[°C] = ([°F] − 32) · 5/9

[°F] = [°C] · 9/5 + 32

Also report the weather conditions as HOT if the temperature exceeds 35°C, WARM if the temperature is between 20°C to 35°C and COLD if the entered temperature is below 20°C.

13. Create a magic square puzzle as per given example below, the sum of all the numbers in a row must be equal, simultaneously the sum of all the numbers in a column must be equal, and the sum of diagonal numbers should also be equal:

- Take the input from the user in all the squares. For a user all the squares will be blank initially and the box given below will contain the text “KEEP TRYING” unless the user enters all correct entries. If user solves the puzzle correctly then a message “WELL DONE” appears below. Format your work accordingly.

	12	3	9	24
	5	8	11	24
	7	13	4	24
24	24	24	24	24

WELL DONE!!!